

PROBABILISTIC SYSTEMS ANALYSIS AND APPLIED PROBABILITY

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LECTURE 1: PROBABILITY MODELS AND AXIOMS

Probability gives us a mathematical way to reason about uncertainty. A **PROBABILITY MODEL** specifies what can happen and assigns probabilities to events. Its two ingredients are a sample space and a probability law.

1.1 Sample spaces: what can happen?

Before assigning probabilities, decide what one complete outcome records. The detail must be enough to answer the questions of interest.

Definition 1.1 [SAMPLE SPACE]. The **SAMPLE SPACE**, denoted by Ω , is the set of all possible outcomes of an experiment. Its outcomes must be **MUTUALLY EXCLUSIVE**: only one outcome occurs in a run. They must also be **COLLECTIVELY EXHAUSTIVE**: every possible result is included.

Example: *Two rolls of a tetrahedral die.* Each roll produces one of 1, 2, 3, 4. Write X for the first result and Y for the second. One complete outcome is the ordered pair (X, Y) , so

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}.$$

There are $4 \times 4 = 16$ outcomes. The order matters: $(1, 2)$ and $(2, 1)$ describe different results. We can display the pairs in a grid or describe the two rolls sequentially with a tree. Each full path through the tree ends at one of the same 16 outcomes.

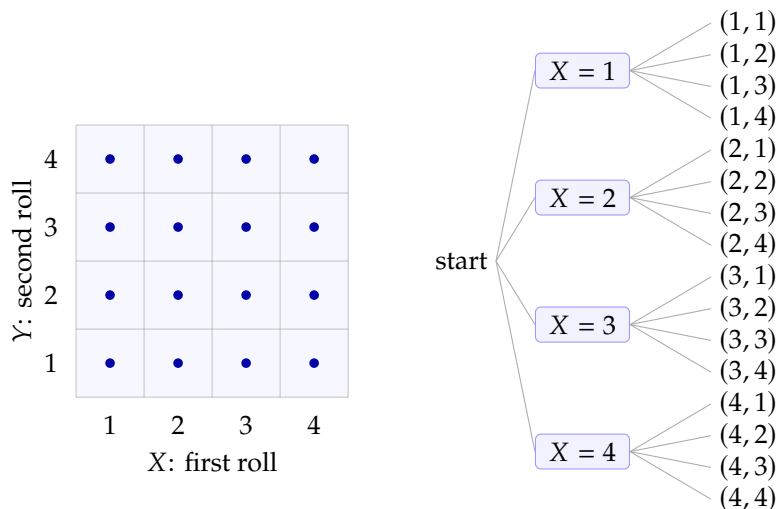


Figure 1: The grid and the tree describe the same sample space. A cell or a complete path records both rolls.

Example: *Two numbers in the unit interval.* If each result can be any real number from 0 to 1, the sample space is

$$\Omega = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\} = [0, 1]^2.$$

It is the entire unit square, including its boundary. The die space is finite; this space is continuous. Both still need a probability law.

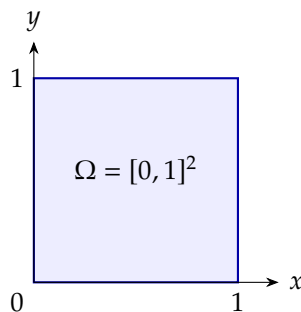


Figure 2: Each point in the unit square is one possible ordered pair.

1.2 Events and probability axioms

We often ask whether a result satisfies a condition, rather than whether one particular outcome occurs.

Definition 1.2 [EVENT AND PROBABILITY LAW]. An event A is a subset of Ω : it occurs when the outcome belongs to A . A PROBABILITY LAW assigns a number $P(A)$ to each event in its domain, subject to the probability axioms below.

For a finite or countably infinite sample space, every subset can be an event. On a continuous space, the domain of the law must be chosen with care; we use ordinary geometric regions here.

Definition 1.3 [PROBABILITY AXIOMS]. For events in the domain of P :

1. NONNEGATIVITY: $P(A) \geq 0$ for every event A .
2. NORMALIZATION: $P(\Omega) = 1$.
3. ADDITIVITY: for DISJOINT events A and B , meaning $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B).$$

The first two axioms say that probabilities cannot be negative and that some outcome in the sample space is certain to occur. The third lets us add probabilities when the events cannot occur together. For overlapping events, this sum would count the common outcomes twice.

Distinct individual outcomes give disjoint singleton events. Repeatedly applying additivity therefore gives

$$P(\{s_1, \dots, s_k\}) = P(\{s_1\}) + \dots + P(\{s_k\}).$$

We may abbreviate $P(\{s\})$ to $P(s)$. The two-event axiom handles finite unions; an infinite union will require a stronger form of additivity.

1.3 A finite probability law: two die rolls

Return to the 16 ordered die outcomes and assign probability $1/16$ to each. The probabilities are nonnegative and sum to $16(1/16) = 1$. An event containing k outcomes has probability $k/16$.

Example: *Either (1, 1) or (1, 2).* These are two distinct outcomes, so

$$P(\{(1, 1), (1, 2)\}) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}.$$

Example: *The first roll is 1.* The second roll can be 1, 2, 3, or 4. Thus

$$\{X = 1\} = \{(1, 1), (1, 2), (1, 3), (1, 4)\}, \quad P(X = 1) = \frac{4}{16} = \frac{1}{4}.$$

The notation $\{X = 1\}$ names the subset of outcomes satisfying the condition.

Example: *The sum is odd.* One roll must be odd and the other even. There are two odd values and two even values, so there are $2 \times 2 = 4$ pairs of each order. Hence

$$P(X + Y \text{ is odd}) = \frac{4 + 4}{16} = \frac{1}{2}.$$

Example: *The smaller roll is 2.* Both rolls must be at least 2, and at least one must equal 2. The event is

$$\{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

There are five outcomes, giving

$$P(\min(X, Y) = 2) = \frac{5}{16}.$$

Counting three pairs with $X = 2$ and three with $Y = 2$ would count $(2, 2)$ twice, so it must be included only once.

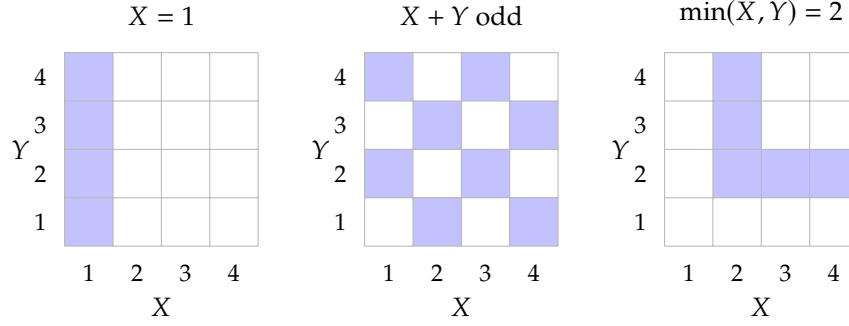


Figure 3: Highlighted cells show the last three die events. Each cell has probability $1/16$. In the last panel, the corner $(2, 2)$ is counted once.

1.4 The discrete uniform law

When a finite sample space has N equally likely outcomes, normalization forces each outcome to have probability $1/N$. This is the **DISCRETE UNIFORM LAW**. Additivity then gives

$$P(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ is the number of outcomes in A and $|\Omega| = N$. Computing a probability becomes a counting problem. Fair coins, fair dice, and well-shuffled decks are modeled this way. The formula relies on the outcomes being equally likely.

1.5 The continuous uniform law

Now give the unit square a **CONTINUOUS UNIFORM LAW**: the probability of a region is its area. The square's total area is 1, so the law is normalized. Equal-area regions have equal probabilities, and areas add for disjoint regions.

Example: *The two numbers sum to at most $1/2$.* The event is

$$A = \{(x, y) \in [0, 1]^2 : x + y \leq \frac{1}{2}\}.$$

It is the triangle bounded by the two coordinate axes and the line $x + y = 1/2$. Both perpendicular sides have length $1/2$, so

$$P(X + Y \leq \frac{1}{2}) = \text{area}(A) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

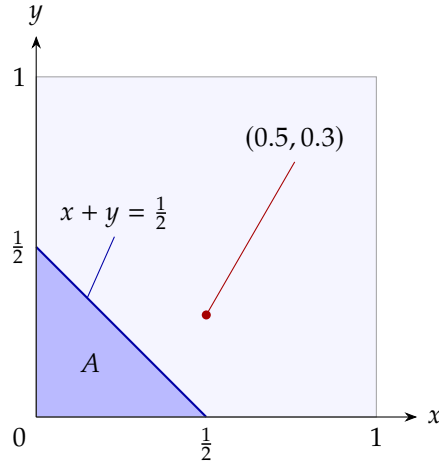


Figure 4: The shaded triangle has area $1/8$. The point $(0.5, 0.3)$ is a possible outcome, but its singleton has zero area.

Example: *The exact pair* $(0.5, 0.3)$. A single point has zero area, so

$$P((X, Y) = (0.5, 0.3)) = 0.$$

The point is still in the sample space. A probability-zero event need not be impossible: in this model every individual point has probability zero, while the whole square has probability one.

1.6 A countably infinite law and countable additivity

Consider the sample space $\Omega = \{1, 2, 3, \dots\}$ with

$$P(n) = 2^{-n}, \quad n = 1, 2, \dots$$

Each probability is positive, and the geometric series checks normalization:

$$\sum_{n=1}^{\infty} 2^{-n} = \frac{1/2}{1 - 1/2} = 1.$$

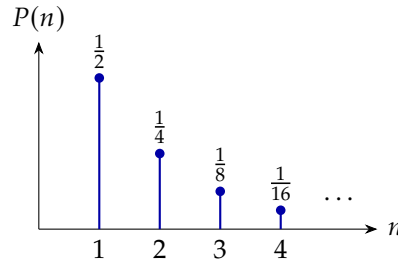


Figure 5: The first four outcome probabilities. Each is half the preceding one, and the pattern continues for every positive integer.

Example: *The outcome is even.* The event $\{2, 4, 6, \dots\}$ contains infinitely many disjoint singleton events. Adding their probabilities gives

$$\begin{aligned} P(\{2, 4, 6, \dots\}) &= \sum_{k=1}^{\infty} 2^{-2k} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \\ &= \frac{1/4}{1 - 1/4} = \frac{1}{3}. \end{aligned}$$

This calculation uses an infinite sum, which finite additivity alone does not justify.

Definition 1.4 [COUNTABLE ADDITIVITY]. If $A_1, A_2, \dots$ are pairwise disjoint events, the COUNTABLE ADDITIVITY axiom requires

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

This is the full additivity axiom for a probability law. It covers a sequence indexed by the positive integers and includes finite additivity as a special case. It justifies the even-outcome calculation above. It does not permit summing singleton probabilities over all the points of the unit square: that collection is uncountable.

Revision. A model needs both a sample space and a probability law. Events are sets of outcomes. Count outcomes under a finite uniform law, measure area under the uniform law on the unit square, and use countable additivity for a countable union of disjoint events.

Common mistakes: using the counting formula without equal likelihood, counting a shared outcome twice, or treating probability zero as impossibility.

Check yourself: Why do two tetrahedral-die rolls give 16 outcomes? Why does $\min(X, Y) = 2$ select five of them? Why can an infinite sum give the probability of an even integer, but cannot add up the probabilities of every point in the unit square?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 1: Probability Models and Axioms](#), PDF pp. 1–3. The calculations are completed and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 2: CONDITIONING AND BAYES' RULE

Learning that an event occurred changes which outcomes remain possible and how their probabilities should be compared. Conditioning makes this update precise. It also gives three useful tools: multiplying probabilities along a sequence, combining disjoint cases, and reversing a conditional probability.

2.1 Start with a probability model

The sample space Ω lists mutually exclusive, collectively exhaustive outcomes at the detail needed for the question. Events are subsets of Ω . Their probabilities are nonnegative, satisfy $P(\Omega) = 1$, and add over disjoint events. For a disjoint sequence $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

To solve a problem, specify the sample space, define its probability law, identify the event of interest, and then calculate. A calculation is only meaningful once these ingredients are clear.

2.2 Conditional probability: restrict and renormalize

Suppose we learn that B occurred. Outcomes outside B can be discarded. Among the outcomes in B , the event A occurs exactly on $A \cap B$. We compare the probability of this intersection with the probability of the new universe B .

Definition 2.1 [CONDITIONAL PROBABILITY]. For events A and B with $P(B) > 0$, the **CONDITIONAL PROBABILITY** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This definition does not assign a value when $P(B) = 0$.

Restricting to B keeps only the outcomes compatible with the information. Dividing by $P(B)$ rescales the remaining probability to one: $P(B \mid B) = P(B)/P(B) = 1$.

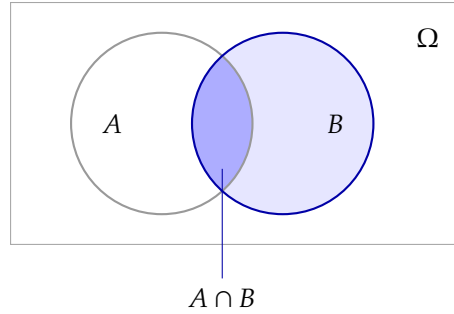


Figure 6: Conditioning keeps B and compares $A \cap B$ with all of B . This is a schematic set diagram; region sizes do not represent probabilities.

Example: *The minimum of two die rolls is 2.* Let X and Y be the two results of a tetrahedral die, with each of the 16 ordered pairs assigned probability $1/16$. Suppose we know

$$B = \{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

Then $P(B) = 5/16$. Each remaining pair has conditional probability $(1/16)/(5/16) = 1/5$.

Let $M = \max(X, Y)$. The event $M = 1$ has no outcome in common with B : if the minimum is 2, the maximum cannot be 1. Therefore

$$P(M = 1 \mid B) = \frac{0}{5/16} = 0.$$

For $M = 2$, the only compatible pair is $(2, 2)$, giving

$$P(M = 2 \mid B) = \frac{1/16}{5/16} = \frac{1}{5}.$$

The denominator counts the five outcomes allowed by the information, not all 16 outcomes in the original space.

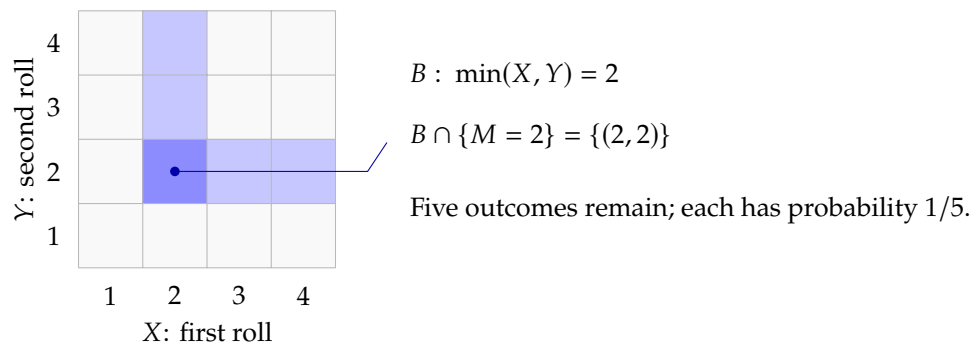


Figure 7: The five colored cells form B . The darker cell $(2, 2)$ is the only one where $M = 2$. Each colored cell has probability $1/5$ given B .

2.3 Building a model from conditional probabilities

A model can be specified in stages. First give the probability of one event, then give the probabilities of later events under each possible earlier result. Write A^c for the COMPLEMENT of A , the event that A does not occur.

Example: *An airplane and a radar reading.* Let A mean that an airplane is flying above, and let B mean that something registers on the radar screen. The model is

$$P(A) = 0.05, \quad P(A^c) = 0.95,$$

with the following conditional probabilities:

$$\begin{aligned} P(B | A) &= 0.99 & P(B^c | A) &= 0.01, \\ P(B | A^c) &= 0.10 & P(B^c | A^c) &= 0.90. \end{aligned}$$

Thus a registration is likely when a plane is present, but registrations also occur when no plane is present. Within each condition, the two radar probabilities sum to one.

Rearranging the conditional-probability definition gives

$$P(A \cap B) = P(A)P(B | A).$$

Multiply the probability of the first stage by the conditional probability of the second stage to obtain the probability of a complete path.

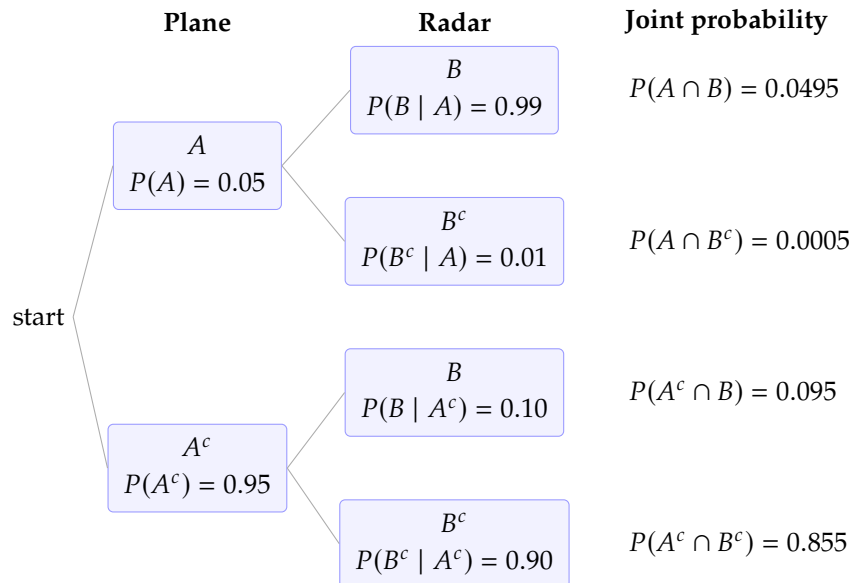


Figure 8: Each path specifies both plane presence and radar status. Multiply the two factors on a path to obtain the joint probability on the right.

For a plane and a registration,

$$P(A \cap B) = 0.05(0.99) = 0.0495.$$

A registration can come from either $A \cap B$ or $A^c \cap B$. These events are disjoint, so

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) \\ &= 0.05(0.99) + 0.95(0.10) = 0.1445. \end{aligned}$$

Once a registration is observed, the probability of a plane is

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.0495}{0.1445} = \frac{99}{289} \approx 0.3426.$$

The two directions answer different questions: $P(B | A) = 0.99$ describes the radar given a plane; $P(A | B) \approx 0.3426$ describes plane presence given a registration. The latter must account for how often planes are present and how often the radar registers without one.

2.4 The multiplication rule

For three events, a complete path specifies A , then B given A , then C given both earlier events. Each factor must use all the information accumulated so far.

Proposition 2.2 [MULTIPLICATION RULE]. *If $P(A) > 0$ and $P(A \cap B) > 0$, then*

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B).$$

The positivity assumptions ensure that the conditional probabilities on the right are defined.

Proof. Apply the conditional-probability definition twice: first condition on the whole event $A \cap B$, then express $P(A \cap B)$ by conditioning on A .

$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap B)P(C | A \cap B) \\ &= P(A)P(B | A)P(C | A \cap B). \end{aligned}$$

□

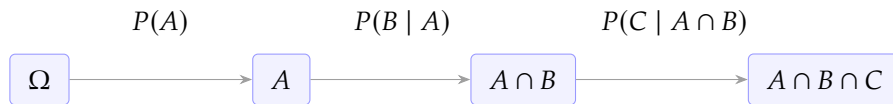


Figure 9: One path through a three-stage model. The last factor conditions on $A \cap B$, because both earlier events have been specified.

2.5 Total probability: combine disjoint cases

Sometimes the probability of B is easier to calculate after dividing the sample space into cases. Compute the contribution from each case, then add the contributions.

Definition 2.3 [PARTITION]. Events A_1, A_2, A_3 form a PARTITION of Ω if they are pairwise disjoint and cover the whole space:

$$A_i \cap A_j = \emptyset \quad (i \neq j), \quad A_1 \cup A_2 \cup A_3 = \Omega.$$

Every outcome belongs to exactly one of the three cases.

Theorem 2.4 [TOTAL PROBABILITY]. For a partition A_1, A_2, A_3 with $P(A_i) > 0$ for each i ,

$$P(B) = P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + P(A_3)P(B \mid A_3).$$

Proof. Each outcome of B belongs to exactly one part of the partition, so $B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)$ is a disjoint union. Additivity gives

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B).$$

For each piece, use $P(A_i \cap B) = P(A_i)P(B \mid A_i)$. Substituting these three products gives the formula. $\square$

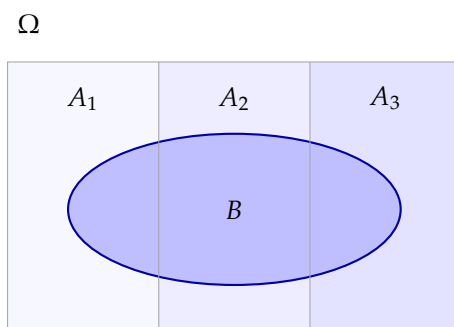


Figure 10: The partition cuts B into three disjoint pieces. Their probabilities add to $P(B)$. The geometry is schematic.

Each term weights a conditional probability by the probability of its case. In the radar example, the two cases are A and A^c , giving the calculation $P(B) = 0.05(0.99) + 0.95(0.10)$.

2.6 Bayes' rule: reverse the conditioning

Suppose we know how likely B is under each case A_i , but observe B and want to infer which case occurred. The PRIOR PROBABILITIES $P(A_i)$ describe the cases before using this observation. The updated probabilities are $P(A_i | B)$.

Theorem 2.5 [BAYES' RULE]. For the partition above, if $P(A_i) > 0$ for each i and $P(B) > 0$, then

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^3 P(A_j)P(B | A_j)}, \quad i = 1, 2, 3.$$

Proof. The conditional-probability definition gives

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}.$$

The multiplication rule supplies the numerator, $P(A_i \cap B) = P(A_i)P(B | A_i)$. The total probability theorem supplies the denominator by adding the contributions from all three cases. Substituting both expressions yields Bayes' rule. $\square$

In [figure 10](#), the numerator is the probability of the piece $A_i \cap B$ and the denominator is the probability of all of B . Thus Bayes' rule compares one case's contribution with the total probability of the observation. In the radar model, it updates $P(A) = 0.05$ to $P(A | B) \approx 0.3426$ after a registration.

Revision. Conditioning restricts the possible outcomes and divides by the probability of the conditioning event. Multiply conditional factors along a path; add disjoint paths for total probability. Bayes' rule divides one case's contribution by the total probability of the observed event.

Common mistakes: interchanging $P(A | B)$ and $P(B | A)$, using a denominator that ignores the information, or forgetting earlier events in the last factor of the multiplication rule.

Check yourself: Why is $P(M = 2 | \min(X, Y) = 2) = 1/5$? Why is the radar model's $P(B) = 0.1445$? Which two ingredients, besides $P(B | A)$, are needed to calculate $P(A | B)$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 2: Conditioning and Bayes' Rule](#), PDF pp. 1–2. The supplied calculations and derivations are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 3: INDEPENDENCE

Independent events do not change one another's probabilities. Independence depends on the probability law: conditioning can remove it or create it.

3.1 Review: probabilities along a coin-toss tree

For $P(B) > 0$, conditioning gives $P(A \mid B) = P(A \cap B)/P(B)$. The multiplication rule, total probability, and Bayes' rule then give

$$\begin{aligned} P(A \cap B) &= P(A)P(B \mid A) = P(B)P(A \mid B), \\ P(B) &= P(A)P(B \mid A) + P(A^c)P(B \mid A^c), \\ P(A \mid B) &= \frac{P(A)P(B \mid A)}{P(B)}. \end{aligned}$$

Each conditional factor requires a conditioning event of positive probability. Multiply along a path and add disjoint paths.

Example: *Three tosses of a biased coin.* At each stage, the probability of heads is p and of tails is $1 - p$, regardless of earlier results. The sample space consists of the eight strings $HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$.

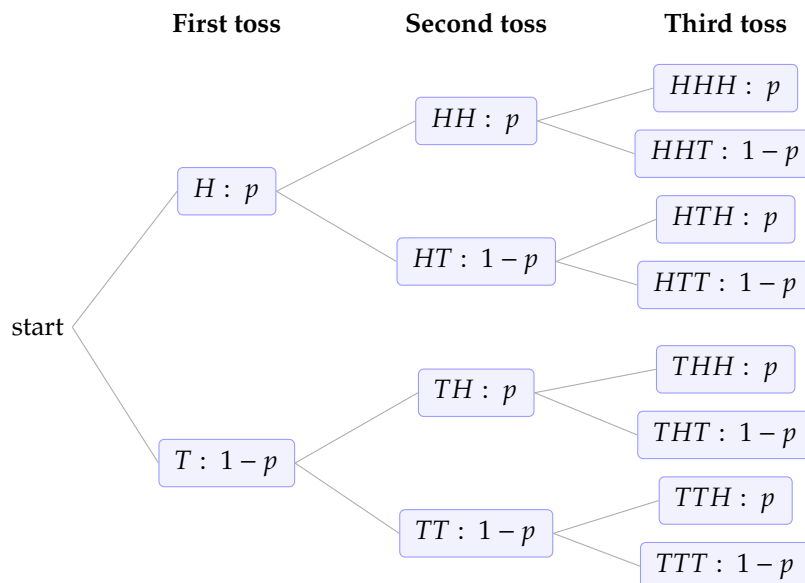


Figure 11: Each box gives the accumulated result and the probability of the newest toss. Multiply the three factors on a path. These factors remain p or $1 - p$ after every earlier history.

The path THT has probability

$$P(THT) = (1 - p)p(1 - p) = p(1 - p)^2.$$

Exactly one head occurs on the three disjoint paths HTT, THT, TTH . Each has the same probability, so

$$P(\text{exactly one head}) = 3p(1 - p)^2.$$

Given exactly one head, the first toss is heads only on HTT . For $0 < p < 1$,

$$P(\text{first toss is } H \mid \text{exactly one head}) = \frac{p(1 - p)^2}{3p(1 - p)^2} = \frac{1}{3}.$$

The bias cancels because all three remaining paths have equal probability. At $p = 0$ or $p = 1$, the conditioning event has probability zero, so this conditional probability is undefined.

3.2 Independence of two events

If learning that A occurred gives no information about B , then $P(B \mid A) = P(B)$, provided $P(A) > 0$. Substitution in the multiplication rule gives a definition that also works for probability-zero events.

Definition 3.1 [INDEPENDENCE]. Events A and B are INDEPENDENT if

$$P(A \cap B) = P(A)P(B).$$

The definition is symmetric in A and B . When $P(A) > 0$, it is equivalent to $P(B \mid A) = P(B)$; when $P(B) > 0$, it is equivalent to $P(A \mid B) = P(A)$.

Proof. If $P(A) > 0$, divide $P(A \cap B) = P(A)P(B)$ by $P(A)$ to obtain $P(B \mid A) = P(B)$. Conversely, multiplying that conditional equality by $P(A)$ gives the defining product. Interchanging A and B proves the other equivalence. $\square$

If $P(A) = 0$, then $A \cap B \subseteq A$ also has probability zero. Thus $P(A \cap B) = 0 = P(A)P(B)$ for every B , even though conditioning on A is undefined. Independence is a numerical property of the probability law; the shape of a set diagram alone does not establish it.

3.3 Conditioning can change independence

After learning that C occurred, use the conditional probability law $P(\cdot | C)$ throughout. Independence in this new universe is tested using conditional probabilities.

Definition 3.2 [CONDITIONAL INDEPENDENCE]. For $P(C) > 0$, events A and B are **CONDITIONALLY INDEPENDENT** given C if

$$P(A \cap B | C) = P(A | C)P(B | C).$$

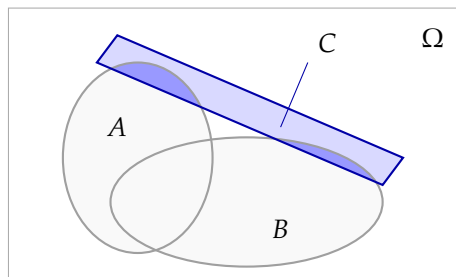


Figure 12: Within C , the pieces $A \cap C$ and $B \cap C$ are disjoint. The diagram is schematic; areas do not specify probabilities.

Suppose A and B are independent before conditioning, but C meets each of them while avoiding $A \cap B$. If both $P(A | C)$ and $P(B | C)$ are positive, then

$$P(A \cap B | C) = 0 \quad \text{while} \quad P(A | C)P(B | C) > 0.$$

They are therefore dependent given C . Unconditional independence does not guarantee conditional independence.

Example: *A coin chosen once, then tossed repeatedly.* Choose Coin A or Coin B with probability $1/2$ each. Coin A has head probability 0.9 ; Coin B has head probability 0.1 . Toss the chosen coin repeatedly, with independent tosses conditional on its identity. Let G be the event that Coin A was chosen, and let H_i mean that toss i is heads. Then

$$P(H_i | G) = 0.9, \quad P(H_i | G^c) = 0.1.$$

Knowing the identity fixes the head probability. Within either conditional model, earlier tosses give no further information about later tosses.

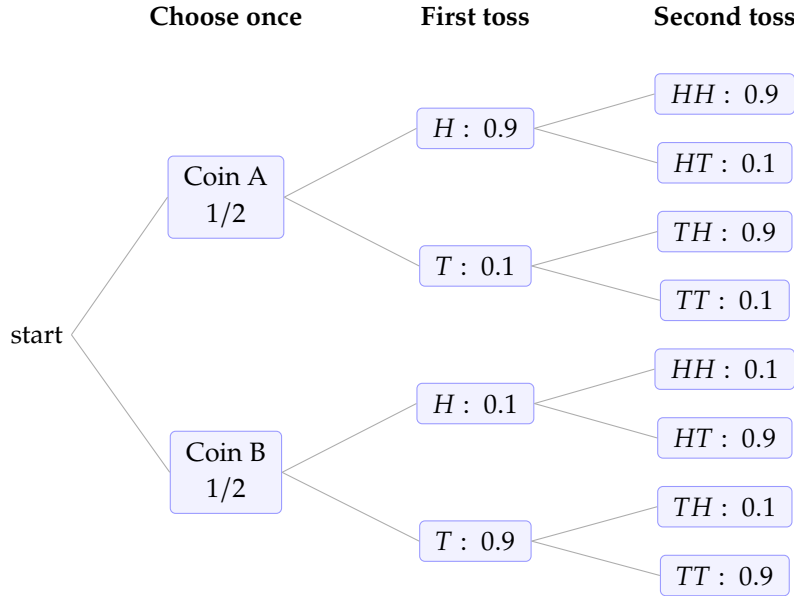


Figure 13: The first choice selects one coin for all tosses. The boxes give the probability of each new step conditional on the preceding history. The two tosses shown are independent given the chosen coin.

Without knowing the coin, total probability gives

$$P(H_{11}) = \frac{1}{2}(0.9) + \frac{1}{2}(0.1) = \frac{1}{2}.$$

Now suppose the first ten tosses were heads. Write $D = H_1 \cap \dots \cap H_{10}$. Conditional independence within each coin model gives $P(D \mid G) = 0.9^{10}$ and $P(D \mid G^c) = 0.1^{10}$. The observation is far more likely with Coin A. Bayes' rule gives

$$P(G \mid D) = \frac{0.9^{10}}{0.9^{10} + 0.1^{10}}.$$

For the next toss, divide the probability of eleven heads by that of ten:

$$\begin{aligned} P(H_{11} \mid D) &= \frac{\frac{1}{2}(0.9^{11}) + \frac{1}{2}(0.1^{11})}{\frac{1}{2}(0.9^{10}) + \frac{1}{2}(0.1^{10})} \\ &= \frac{0.9^{11} + 0.1^{11}}{0.9^{10} + 0.1^{10}} \approx 0.9. \end{aligned}$$

This differs from $P(H_{11}) = 1/2$. Earlier results reveal information about the shared coin and change the probability of later results. Thus conditional independence does not guarantee unconditional independence.

3.4 Independence of a collection of events

For a collection, the idea is stronger than checking two events at a time: information about any group of events must give no information about the remaining events.

Definition 3.3 [INDEPENDENCE OF A COLLECTION]. Events $A_1, \dots, A_n$ are **MUTUALLY INDEPENDENT** if, for every nonempty subset $S \subseteq \{1, \dots, n\}$,

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i).$$

Here $\prod$ means multiplication over the indicated indices. The condition must hold for every subcollection, not just all n events together.

For three events, this requires the three pairwise products and also

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3).$$

For a larger independent collection, even combinations of events and their complements involving disjoint groups of indices give no information about one another. For example, when $n \geq 6$ and $P(A_5 \cap A_6^c) > 0$,

$$P(A_1 \cap (A_2^c \cup A_3) \mid A_5 \cap A_6^c) = P(A_1 \cap (A_2^c \cup A_3)).$$

The condition concerns indices 5, 6, while the event of interest concerns indices 1, 2, 3.

3.5 Pairwise independence is not enough

Events are **PAIRWISE INDEPENDENT** if each pair of distinct events is independent. This checks only subcollections of size two. It need not establish mutual independence.

Example: *Two fair coin tosses and whether they match.* Consider two independent fair tosses, with the four equally likely outcomes HH, HT, TH, TT . Define

$$A = \{HH, HT\}, \quad B = \{HH, TH\}, \quad C = \{HH, TT\}.$$

Thus A means the first toss is heads, B the second is heads, and C that both results are the same.

		Second toss		
		H	T	
First toss	H	HH	HT	
	T	TH	TT	

A: first toss is H
 B: second toss is H
 C: the tosses match

Figure 14: Every cell has probability $1/4$. The two colored cells form C . Knowing both A and B identifies HH , which is in C .

Each event contains two outcomes, so $P(A) = P(B) = P(C) = 1/2$. Each pairwise intersection is the singleton $\{HH\}$:

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}.$$

All three pairs are independent. But the triple intersection is also $\{HH\}$, giving

$$P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C).$$

Equivalently,

$$P(C \mid A \cap B) = \frac{1/4}{1/4} = 1 \neq \frac{1}{2} = P(C).$$

Neither toss alone changes the probability of matching; knowing both tosses determines whether they match. Pairwise independence misses this dependence on combined information.

3.6 The king's sibling: specify the experiment

Example: *A king comes from a family with two children. What is the probability that his sibling is female?* Assume the parents decided in advance to have exactly two children. Each child is a boy or girl with probability $1/2$, independently of the other. Assume that a boy takes precedence in succession, so a king exists exactly when at least one child is a boy.

Order the children by birth. The four outcomes BB, BG, GB, GG each have probability $1/4$, where B means boy and G girl. The information that there is a king is the event

$$K = \{BB, BG, GB\}, \quad P(K) = \frac{3}{4}.$$

His sibling is female in BG or GB . Writing F for this event within K gives

$$P(F \mid K) = \frac{P(F \cap K)}{P(K)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

		Younger child		
		B	G	
Older child	B	BB	BG	
	G	GB	GG excluded	

K : at least one boy
 $F \cap K$: BG or GB
 $P(F | K) = 2/3$

Figure 15: Under the fixed two-child model, conditioning on K removes GG . The other three outcomes are equally likely; two give a female sibling. The darker cells are $F \cap K$.

The answer depends on the experiment. If the parents instead have children until the first boy and then stop, having exactly two children forces the outcome GB . Under that rule, the king's sibling is female with probability one. The phrase "a family with two children" alone does not specify how the family came to have two children.

Revision. Independence is the product condition $P(A \cap B) = P(A)P(B)$. Conditional independence uses the same condition under $P(\cdot | C)$. Mutual independence checks every subcollection; pairwise independence checks only pairs.

Common mistakes: carrying independence into a conditional model without checking it, inferring mutual independence from pairwise products, or leaving the experiment's selection and stopping rules unspecified.

Check yourself: Why does exactly one head give a first-toss head probability of $1/3$, regardless of the bias? Why do ten heads change the next-toss probability for the unknown coin? Which product fails in the matching-toss example? What assumptions produce the sibling probability $2/3$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 3: Independence](#), PDF pp. 1–2, and the [accompanying transcript](#) (conditional independence and the sibling model, especially 41:29–46:28). The supplied calculations and derivations are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 4: COUNTING

Counting turns a probability question into two questions: how many outcomes are possible, and how many satisfy the event? The main task is to describe each outcome exactly once, taking care about order, repetition, and which groups are labeled.

4.1 The discrete uniform law

If a finite sample space Ω has equally likely outcomes, each outcome has probability $1/|\Omega|$. For an event $A \subseteq \Omega$,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Here $|A|$ is the number of outcomes in A . Count the numerator and denominator using the same kind of outcome. The formula requires equal likelihood; counting alone does not establish that assumption.

4.2 The basic counting principle

Construct an outcome through successive choices. If every partial outcome has the same number of extensions at a given stage, the total can be found by multiplying the numbers of choices.

Proposition 4.1 [BASIC COUNTING PRINCIPLE]. *Suppose an outcome is constructed in r stages. At stage i , there are n_i choices for every permitted history of earlier stages. If each complete sequence of choices describes exactly one outcome, and each outcome has exactly one such description, the number of outcomes is*

$$n_1 n_2 \cdots n_r.$$

Proof. After the first stage there are n_1 partial outcomes. Each has n_2 extensions, giving $n_1 n_2$ partial outcomes after stage two. Repeating this reasoning at each stage gives the product. The unique-description assumption prevents counting the same outcome more than once. $\square$

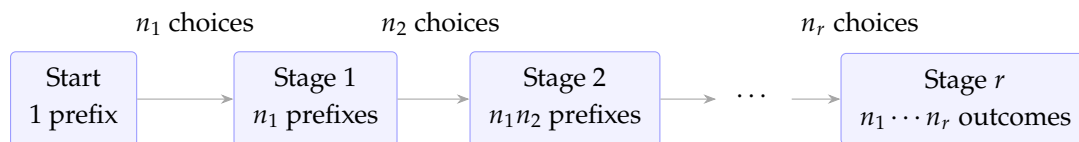


Figure 16: Each stage extends every existing prefix by the indicated number of choices. The boxes count prefixes, not probabilities.

Example: License plates with three letters followed by four digits. Use 26 letters and the ten digits 0, . . . , 9 in these fixed positions. With repetition allowed, the number of plates is

$$26^3 10^4 = 175,760,000.$$

If letters cannot repeat and digits cannot repeat, the choices decrease as positions are filled:

$$(26 \cdot 25 \cdot 24)(10 \cdot 9 \cdot 8 \cdot 7) = 78,624,000.$$

	Three letters			Four digits			
Repetition allowed	26	26	26	10	10	10	10
No repetition	26	25	24	10	9	8	7

Figure 17: Numbers in the slots count the available choices. Multiply across a row. Without repetition, used symbols are unavailable later.

A **PERMUTATION** of n distinct elements is an ordering of all of them. There are n choices for the first position, $n - 1$ for the second, and so on, giving

$$n(n - 1) \cdots 2 \cdot 1 = n!.$$

The notation $n!$ is the **FACTORIAL** of n ; by convention, $0! = 1$.

To construct a subset of $\{1, \dots, n\}$, decide separately for each element whether to include it. There are two choices at each of n stages, so the number of subsets, including the empty set, is

$$2^n.$$

Example: *Six die rolls, all different.* Roll a fair six-sided die independently six times. An outcome is an ordered sequence of six results. All 6^6 sequences are equally likely. For the results to be different, the numbers of choices are 6, 5, 4, 3, 2, 1, giving $6!$ favorable sequences. Thus

$$P(\text{all six different}) = \frac{6!}{6^6} = \frac{720}{46,656} = \frac{5}{324} \approx 0.01543.$$

The denominator allows repeated results; the numerator forbids them.

4.3 From ordered choices to combinations

Choosing k distinct elements in order from an n -element set is a k -PERMUTATION. For integers $0 \leq k \leq n$, its count is

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

For $k = 0$, there is one empty sequence. An unordered selection must identify all the different orderings of the same chosen elements.

Definition 4.2 [COMBINATIONS]. The BINOMIAL COEFFICIENT $\binom{n}{k}$, read “ n choose k ,” is the number of k -element subsets of a given n -element set. These unordered selections are called COMBINATIONS.

Proposition 4.3 [COUNTING COMBINATIONS]. For integers $0 \leq k \leq n$,

$$\boxed{\binom{n}{k} = \frac{n!}{k!(n-k)!}}.$$

Proof. Count ordered selections of k distinct elements in two ways. Choosing the elements one at a time gives $n!/(n-k)!$ selections. Alternatively, choose a k -element subset in $\binom{n}{k}$ ways, then order its elements in $k!$ ways. Both procedures describe the same ordered selections, so

$$\binom{n}{k} k! = \frac{n!}{(n-k)!}.$$

Dividing by $k!$ gives the formula. □

The factor $k!$ removes repeated counts of the same subset. Also, every subset has one of the sizes $0, 1, \dots, n$. Counting subsets by size or by the include/exclude decisions above gives

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

4.4 Binomial probabilities

Consider n independent tosses of a coin with head probability p . A specified sequence with k heads and $n - k$ tails has probability $p^k(1-p)^{n-k}$. For the sequence $HTTHHH$, for example,

$$P(HTTHHH) = p(1-p)(1-p)ppp = p^4(1-p)^2.$$

The positions of the heads identify the sequence. Choosing k of the n positions therefore counts the sequences with exactly k heads.

Proposition 4.4 [BINOMIAL PROBABILITY]. For $0 < p < 1$ and $k = 0, 1, \dots, n$,

$$P(\text{exactly } k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Proof. Exactly k heads is a disjoint union of the $\binom{n}{k}$ sequences whose head positions form a k -element subset. Independence gives each sequence probability $p^k (1-p)^{n-k}$. Add their equal probabilities to obtain the formula. $\square$

If $p = 0$, the all-tail sequence has probability one; if $p = 1$, the all-head sequence has probability one. These are the boundary cases of the same model. For a biased coin, sequences with different head counts need not be equally likely. Equal likelihood is being used only among sequences with a fixed head count.

4.5 Conditioning on the number of heads

Example: *Three heads in ten tosses.* Toss a coin independently ten times, with $0 < p < 1$. Let B be the event that exactly three tosses are heads, and let A be the event that the first two tosses are heads. Find $P(A | B)$.

Every sequence in B has probability $p^3 (1-p)^7$. After conditioning on B , these sequences remain equally likely, even when the coin is biased. There are

$$|B| = \binom{10}{3} = 120$$

possible sets of head positions. To be in $A \cap B$, positions 1 and 2 must be heads, leaving exactly one head to place among positions 3 through 10. Hence

$$|A \cap B| = \binom{8}{1} = 8.$$

In B : choose three of all ten positions

In $A \cap B$: the first two heads are fixed

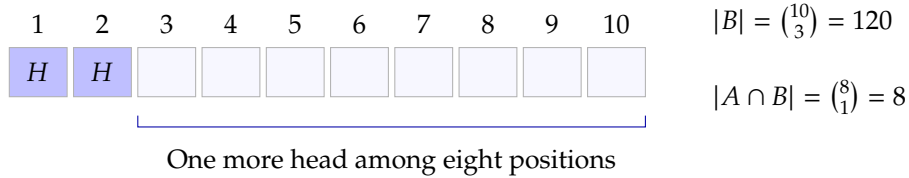


Figure 18: Fixing the first two heads leaves eight possible positions for the last head. Without those fixed heads, any three of the ten positions may be the head positions.

The conditional uniform law gives

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{120} = \frac{1}{15}.$$

Equivalently, the common factor $p^3(1-p)^7$ cancels from the conditional-probability ratio. The denominator counts only sequences compatible with B .

4.6 Partitions: dealing a deck to four players

Example: *Each player receives an ace.* Deal a uniformly shuffled 52-card deck to four labeled players, with 13 cards per player. What is the probability that each receives one ace? An outcome records each player's hand, with order within a hand ignored. The four hands are disjoint and cover the deck, so they form a partition.

To count all outcomes, order the 52 distinct cards and divide the list into four consecutive blocks of 13 cards. Each fixed deal appears $(13!)^4$ times, because cards within any hand can be reordered without changing the deal. Thus

$$|\Omega| = \frac{52!}{(13!)^4}.$$

Every partition has the same number of full-deck orderings, so these outcomes are equally likely. Do not divide by $4!$: the players are labeled, and exchanging their hands changes the outcome.

For a favorable deal, first give the four distinct aces to the four players, one each. There are $4! = 4 \cdot 3 \cdot 2 \cdot 1$ assignments. Then distribute the 48 non-aces, giving 12 to each player. The same ordering-and-grouping argument gives $48!/(12!)^4$ assignments of the non-aces. Therefore

$$|A| = 4! \frac{48!}{(12!)^4},$$

where A is the event that each player has an ace.

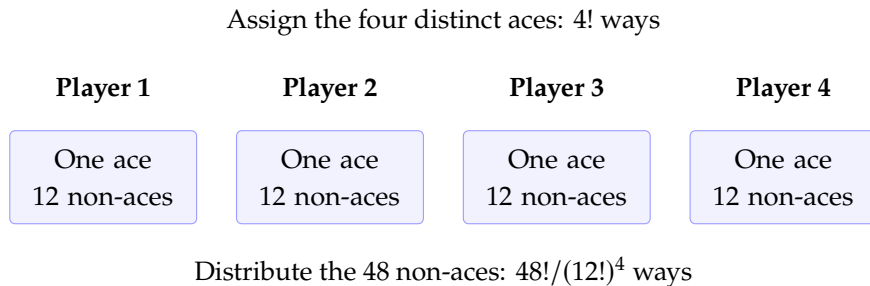


Figure 19: A favorable deal puts one distinct ace and 12 non-aces in each labeled hand. Choose the ace assignment, then the distribution of the remaining cards.

Taking the ratio of favorable to total outcomes and canceling factorials gives

$$\begin{aligned} P(A) &= \frac{4! 48! / (12!)^4}{52! / (13!)^4} \\ &= \frac{4! 13^4}{52 \cdot 51 \cdot 50 \cdot 49} = \frac{2197}{20825} \approx 0.10550. \end{aligned}$$

The numerator and denominator both count unordered hands assigned to labeled players.

Revision. Multiply choices across stages when each stage has a fixed number of extensions. Ordered selections of k distinct items have count $n!/(n-k)!$; unordered selections have count $\binom{n}{k}$. For independent coin tosses, exactly k heads has probability $\binom{n}{k} p^k (1-p)^{n-k}$.

Common mistakes: using a uniform counting ratio without equal likelihood, mixing ordered and unordered outcomes, allowing repetition in only one of the two counts, or dividing by permutations of labeled players.

Check yourself: Why does counting combinations require division by $k!$? Why does the bias cancel after conditioning on three heads? Why is the deck count divided by $(13!)^4$, but not by $4!$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 4: Counting](#), PDF pp. 1–2. The supplied calculations and counting arguments are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 5: DISCRETE RANDOM VARIABLES, PMFs, AND EXPECTATIONS

A probability model describes outcomes. A random variable records a number from each outcome. Its probability mass function collects the probability attached to each number; expectation describes the center of this distribution, and variance describes its spread.

5.1 Random variables: numbers assigned to outcomes

Definition 5.1 [RANDOM VARIABLE]. A RANDOM VARIABLE is a function $X : \Omega \rightarrow \mathbb{R}$. For each outcome ω , the number $X(\omega)$ is the value recorded by X .

The rule X is fixed; the outcome determines which value it takes. Use a capital letter X for the random variable and a lower-case letter x for a possible numerical value. Several random variables can record different numbers from the same outcome.

A random variable is DISCRETE when its possible values form a finite or countably infinite set. Random variables can also take continuous values, but the probability mass functions below describe the discrete case.

5.2 Probability mass functions

To find the probability of a value x , collect the outcomes that produce that value. The event $\{X = x\}$ is the set $\{\omega \in \Omega : X(\omega) = x\}$.

Definition 5.2 [PROBABILITY MASS FUNCTION]. The PROBABILITY MASS FUNCTION, or PMF, of a discrete random variable X is

$$p_X(x) = P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\}).$$

It describes the PROBABILITY DISTRIBUTION of X .

Each value has nonnegative probability, and the probabilities of all possible values add to one:

$$p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

The sum runs over the possible values of X , rather than over individual outcomes. The events $\{X = x\}$ are disjoint and cover the sample space, which explains normalization.

Example: *Tosses until the first head.* Toss a coin independently, with head probability $p > 0$. Let X be the number of tosses up to and including the first head. For $X = k$, the first $k - 1$ tosses must be tails and the k th must be heads. For $0 < p < 1$,

$$p_X(k) = P(\underbrace{TT \cdots T}_{k-1 \text{ tails}} H) = (1 - p)^{k-1} p, \quad k = 1, 2, \dots$$

This is the GEOMETRIC PMF. Its probabilities sum to $p \sum_{j=0}^{\infty} (1 - p)^j = p/[1 - (1 - p)] = 1$. Equivalently, the probability of seeing no head in the first m tosses is $(1 - p)^m$, which tends to zero. A head therefore occurs eventually with probability one. If $p = 1$, then $X = 1$ with probability one.

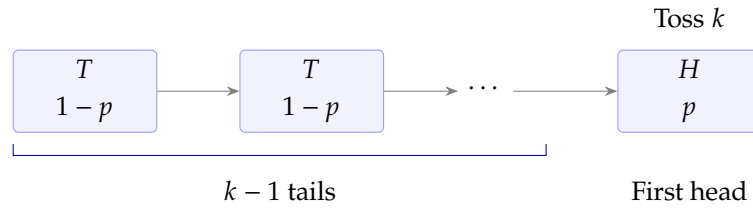


Figure 20: For the event $X = k$, every earlier toss is a tail and the last toss is a head. Independence lets us multiply the probabilities of these required results.

5.3 Computing a PMF by grouping outcomes

For each possible value x , collect all outcomes on which $X = x$ and add their probabilities. Repeat for the other values. Different outcomes can contribute to the same PMF value.

Example: *The minimum of two die rolls.* Roll a fair tetrahedral die twice, independently. Let F and S be the first and second results, and set $X = \min(F, S)$. The 16 ordered pairs (f, s) are equally likely, each with probability $1/16$.

For $X = 2$, both results must be at least 2 and at least one must equal 2. The five pairs are

$$(2, 2), (2, 3), (2, 4), (3, 2), (4, 2),$$

so $p_X(2) = 5/16$. Count $(2, 2)$ once, even though both rolls equal 2. Grouping the other pairs by their minimum gives the complete PMF:

x	1	2	3	4
$p_X(x)$	7/16	5/16	3/16	1/16

These four probabilities sum to one. The values of X are not equally likely, even though the underlying die outcomes are.

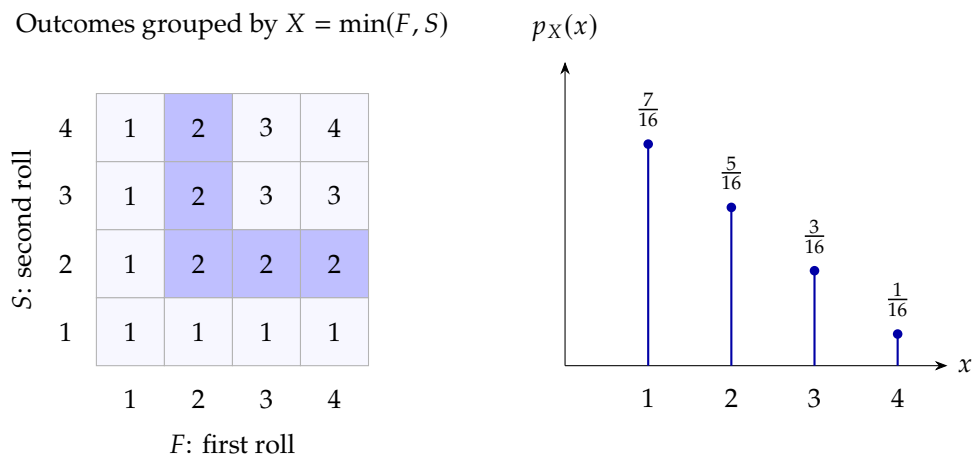


Figure 21: Each grid cell shows $X = \min(F, S)$ and has probability $1/16$. The highlighted cells give $X = 2$. Grouping cells with the same number produces the PMF on the right.

5.4 The binomial PMF

Let X count the heads in n independent coin tosses, with head probability p . A sequence with k heads and $n - k$ tails has probability $p^k(1 - p)^{n-k}$. There are $\binom{n}{k}$ such sequences, one for each choice of the k head positions. Thus the BINOMIAL PMF is

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n, \quad 0 < p < 1.$$

The PMF is zero at other values. At $p = 0$, $X = 0$ with probability one; at $p = 1$, $X = n$ with probability one.

Example: *Two heads in four tosses.* For $n = 4$, the event $X = 2$ consists of six disjoint sequences:

$$HHTT, HTHT, HTTH, THHT, THTH, TTHH.$$

Each has probability $p^2(1 - p)^2$. Adding them gives

$$p_X(2) = 6p^2(1 - p)^2 = \binom{4}{2} p^2(1 - p)^2.$$

The coefficient counts sequences, and the remaining factor gives the probability of one such sequence.

5.5 Expectation: a probability-weighted average

Values with more probability receive more weight in the average.

Definition 5.3 [EXPECTATION]. For a discrete random variable X , its EXPECTATION, OR MEAN, is

$$\mathbb{E}[X] = \sum_x x p_X(x),$$

provided $\sum_x |x| p_X(x) < \infty$.

For finitely many possible values, the sum is automatically finite. For infinite sets of values, the expectation identities below assume that the relevant sums are finite. Think of $\mathbb{E}[X]$ as the center of gravity when mass $p_X(x)$ is placed at position x . It also represents the average approached over many independent repetitions of the same experiment; this interpretation requires a limit result.

Example: *Uniform on $0, 1, \dots, n$.* Let n be a nonnegative integer. Each value has probability $1/(n+1)$, so

$$\mathbb{E}[X] = \frac{0 + 1 + \dots + n}{n+1}.$$

To evaluate the numerator, write $S = 0 + 1 + \dots + n$ and also $S = n + (n-1) + \dots + 0$. Adding the two expressions gives $2S = n(n+1)$. Therefore

$$\mathbb{E}[X] = \frac{n(n+1)/2}{n+1} = \left\lfloor \frac{n}{2} \right\rfloor.$$

Equal probabilities at positions equally far from $n/2$ also explain why this is the center.

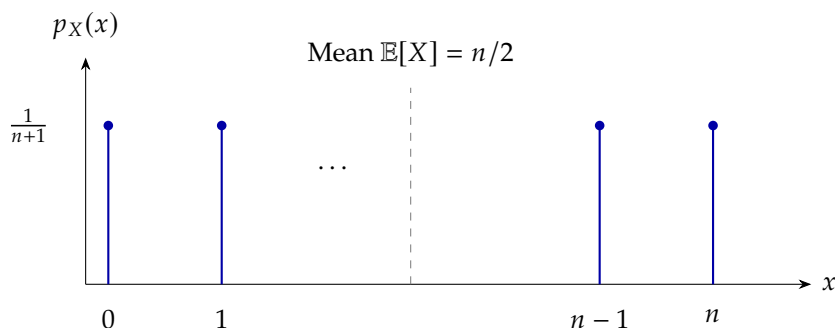


Figure 22: The uniform PMF assigns equal mass to all $n+1$ positions. Its center is $n/2$. The spacing is schematic; omitted positions are indicated by dots.

5.6 Expectations of functions and affine expressions

Suppose $Y = g(X)$. One way to calculate $\mathbb{E}[Y]$ is to find the PMF of Y and sum $y p_Y(y)$. We can instead use the PMF of X directly.

Proposition 5.4 [EXPECTATION OF A FUNCTION]. *If $\sum_x |g(x)|p_X(x) < \infty$, then*

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x).$$

Proof. Group the values of X according to the common value they produce under g . For a possible value y of Y ,

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x).$$

Substitute this into the definition of expectation:

$$\begin{aligned} \mathbb{E}[Y] &= \sum_y y \sum_{x:g(x)=y} p_X(x) \\ &= \sum_y \sum_{x:g(x)=y} g(x)p_X(x) = \sum_x g(x)p_X(x). \end{aligned}$$

Every value of X belongs to one group. Absolute convergence justifies regrouping an infinite sum. $\square$

In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. Taking a function of an average need not give the average of that function. An **AFFINE FUNCTION** has the form $g(x) = \alpha x + \beta$, with constants α, β . For these functions, the two operations agree.

Proposition 5.5 [CONSTANTS AND AFFINE FUNCTIONS]. *For constants α, β and a random variable with finite mean,*

$$\mathbb{E}[\alpha] = \alpha, \quad \mathbb{E}[\alpha X] = \alpha \mathbb{E}[X], \quad \boxed{\mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta}.$$

Proof. The PMF sums to one, so a constant contributes $\sum_x \alpha p_X(x) = \alpha$. Applying the expectation-of-a-function formula to $g(x) = \alpha x + \beta$ gives

$$\sum_x (\alpha x + \beta)p_X(x) = \alpha \sum_x x p_X(x) + \beta \sum_x p_X(x) = \alpha \mathbb{E}[X] + \beta.$$

Taking $\beta = 0$ gives the scaling identity. $\square$

5.7 Variance: squared distance from the mean

The mean locates the center, but does not describe how far values lie from it. Squaring their deviations from the mean prevents positive and negative deviations from canceling.

Definition 5.6 [SECOND MOMENT AND VARIANCE]. The SECOND MOMENT of X is $\mathbb{E}[X^2] = \sum_x x^2 p_X(x)$. When this is finite, write $\mu = \mathbb{E}[X]$. The VARIANCE of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x).$$

The expectation-of-a-function formula applies with $g(x) = (x - \mu)^2$. It also gives a useful way to compute variance.

Proposition 5.7 [VARIANCE IDENTITIES]. For finite second moment,

$$\boxed{\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}, \quad \text{Var}(X) \geq 0.$$

For constants α, β ,

$$\boxed{\text{Var}(\alpha X + \beta) = \alpha^2 \text{Var}(X)}.$$

Proof. Expand the square inside the defining sum and use $\sum_x x p_X(x) = \mu$ and $\sum_x p_X(x) = 1$:

$$\begin{aligned} \text{Var}(X) &= \sum_x (x^2 - 2\mu x + \mu^2) p_X(x) \\ &= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2. \end{aligned}$$

Nonnegativity follows from the defining sum of nonnegative squared deviations, not from treating the last difference as automatically nonnegative.

For $Y = \alpha X + \beta$, its mean is $\mathbb{E}[Y] = \alpha\mu + \beta$. Consequently

$$Y - \mathbb{E}[Y] = \alpha(X - \mu), \quad \text{Var}(Y) = \mathbb{E}[\alpha^2(X - \mu)^2] = \alpha^2 \text{Var}(X).$$

□

A shift β moves the values and their mean by the same amount, leaving deviations unchanged. Scaling by α multiplies each deviation by α , so squared deviations scale by α^2 .

Revision. A random variable assigns a number to each outcome. Its PMF groups outcomes by that number and adds their probabilities. Expectation is the probability-weighted average; a function's expectation can be computed directly from the original PMF. Variance is the average squared deviation from the mean, equivalently $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Common mistakes: confusing an outcome with a value of X , assuming equally likely outcomes give equally likely values, replacing $\mathbb{E}[g(X)]$ by $g(\mathbb{E}[X])$ for an arbitrary function, or scaling variance by α instead of α^2 .

Check yourself: Which five die outcomes give $X = 2$? Why does the geometric PMF contain $k - 1$ tail factors? Why does adding a constant change the mean but leave variance unchanged?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 5: Discrete Random Variables; Probability Mass Functions; Expectations](#), PDF pp. 1–2. The supplied calculations and expectation and variance arguments are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 6: DISCRETE RANDOM VARIABLE EXAMPLES AND JOINT PMFs

Expectations describe averages, but a function of an average need not equal the average of that function. Conditioning changes the probabilities used in an average. A joint PMF records how two random variables behave together.

6.1 Review: means, deviations, and standard deviation

A random variable X assigns a real number to each outcome. For a discrete X , its PMF is $p_X(x) = P(X = x)$. With the relevant sums finite,

$$\mathbb{E}[X] = \sum_x x p_X(x), \quad \mathbb{E}[g(X)] = \sum_x g(x) p_X(x), \quad \mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta.$$

Writing $\mu = \mathbb{E}[X]$, the deviations from the mean average to zero:

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0.$$

Their squares need not average to zero. They give the variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x) = \mathbb{E}[X^2] - \mu^2.$$

Definition 6.1 [STANDARD DEVIATION]. The STANDARD DEVIATION of X is

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Variance has squared units; standard deviation has the same units as X .

6.2 Random speed and average travel time

Travel a distance of 200 miles at a speed V that is constant during the trip but random before it begins. The speed is either 1 or 200 miles per hour, each with probability $1/2$.

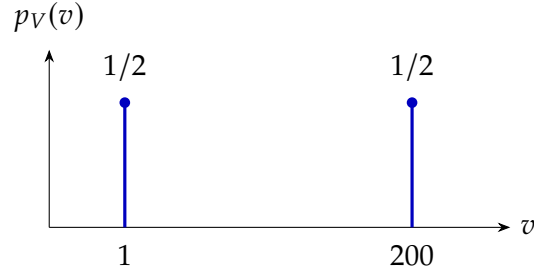


Figure 23: Both speeds have probability $1/2$. The horizontal spacing is schematic; the two values are 1 and 200 miles per hour.

The mean speed is

$$\mathbb{E}[V] = \frac{1}{2}(1) + \frac{1}{2}(200) = \frac{201}{2} = 100.5 \text{ miles per hour.}$$

Both possible speeds are $199/2$ from this mean, so

$$\text{Var}(V) = \frac{1}{2} \left(-\frac{199}{2} \right)^2 + \frac{1}{2} \left(\frac{199}{2} \right)^2 = \frac{39601}{4}, \quad \sigma_V = \frac{199}{2} = 99.5 \text{ miles per hour.}$$

Travel time, in hours, is the function $T = t(V) = 200/V$:

V	1	200
$T = 200/V$	200	1
Probability	$1/2$	$1/2$

Average the times using their probabilities:

$$\mathbb{E}[T] = \sum_v \frac{200}{v} p_V(v) = \frac{1}{2}(200) + \frac{1}{2}(1) = \frac{201}{2} = 100.5 \text{ hours.}$$

Using the mean speed instead would give

$$\frac{200}{\mathbb{E}[V]} = \frac{400}{201} \text{ hours,} \quad \mathbb{E}[200/V] \neq \frac{200}{\mathbb{E}[V]}.$$

The slow trip contributes 200 hours to the possible times, so it strongly affects their average. Taking a reciprocal does not commute with averaging.

Also, $TV = 200$ on every outcome, hence

$$\mathbb{E}[TV] = 200 \neq \mathbb{E}[T]\mathbb{E}[V] = \frac{40401}{4}.$$

The speed and time come from the same trip. Their product must be averaged using the values that occur together.

6.3 Conditional PMFs and conditional expectations

Once we know that an event A occurred, use conditional probabilities as the new weights. Assume $P(A) > 0$.

Definition 6.2 [CONDITIONAL PMF AND EXPECTATION]. The CONDITIONAL PMF of X given A is

$$p_{X|A}(x) = P(X = x | A) = \frac{P(\{X = x\} \cap A)}{P(A)}.$$

Its probabilities sum to one. The CONDITIONAL EXPECTATION is

$$\mathbb{E}[X | A] = \sum_x x p_{X|A}(x),$$

when the corresponding absolute-value sum is finite.

Example: *A uniform variable after conditioning.* Let X be uniform on $\{1, 2, 3, 4\}$, so $p_X(x) = 1/4$ at these four values. Take $A = \{X \geq 2\}$. Then $P(A) = 3/4$. Conditioning removes the value 1 and divides each remaining probability by $3/4$:

$$p_{X|A}(x) = \begin{cases} 1/3, & x \in \{2, 3, 4\}, \\ 0, & \text{otherwise.} \end{cases} \quad \mathbb{E}[X | A] = \frac{2 + 3 + 4}{3} = 3.$$

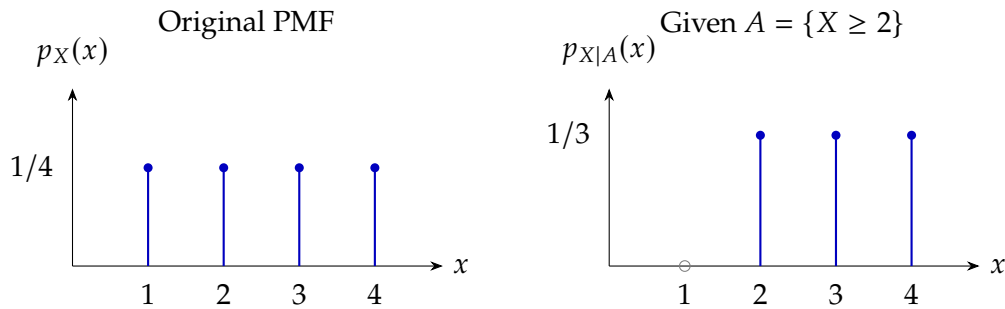


Figure 24: Conditioning on $X \geq 2$ removes the probability at 1. The retained probabilities are rescaled to sum to one.

6.4 The geometric PMF and memorylessness

Toss a coin independently until the first head, with head probability $0 < p < 1$. Let X count all tosses, including the head, and write $q = 1 - p$. The first $k - 1$ tosses must be tails, giving

$$p_X(k) = q^{k-1}p, \quad k = 1, 2, \dots, \quad \mathbb{E}[X] = \sum_{k=1}^{\infty} k q^{k-1} p.$$

Suppose we already know that $X > 2$. The first two tosses were tails, with probability q^2 . For $k \geq 3$,

$$p_{X|X>2}(k) = \frac{q^{k-1}p}{q^2} = q^{k-3}p.$$

$$P(R = j \mid X > 2) = P(X = j + 2 \mid X > 2) = q^{j-1}p.$$

Thus, given that the first two tosses failed, the remaining waiting time has the original geometric PMF. This is the **MEMORYLESS PROPERTY**. The total waiting time still starts at 3 under this condition; it is the remaining waiting time that starts at 1.

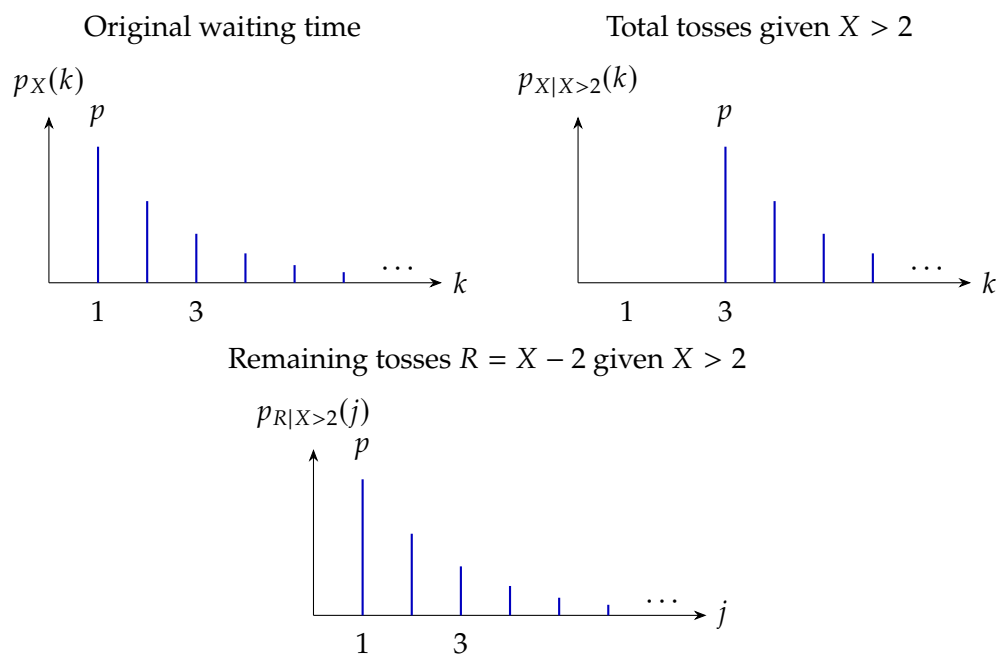


Figure 25: The geometric PMFs are schematic. Conditioning on two tails renormalizes the tail of the original PMF. Subtracting the two elapsed tosses restores its original shape and starting value.

6.5 Total expectation and the geometric mean

Split the sample space into disjoint events $A_1, \dots, A_n$ whose union is Ω . First average within each event, then weight these averages by how likely the events are. Events of probability zero can be omitted.

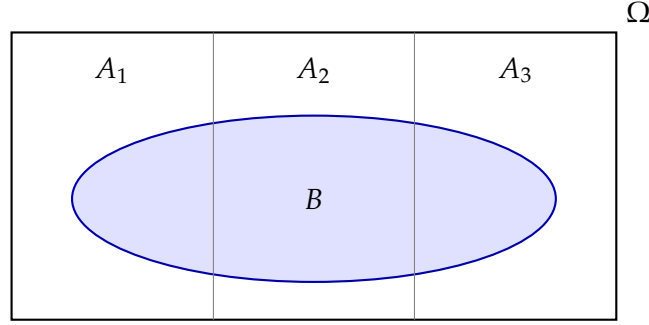


Figure 26: The events A_1, A_2, A_3 partition Ω . The portions of B in these events are disjoint and together make up B .

For any event B , total probability gives

$$P(B) = \sum_{i=1}^n P(A_i)P(B | A_i).$$

Taking $B = \{X = x\}$ yields

$$p_X(x) = \sum_{i=1}^n P(A_i)p_{X|A_i}(x).$$

Theorem 6.3 [TOTAL EXPECTATION]. *For a discrete random variable with finite absolute expectation,*

$$\mathbb{E}[X] = \sum_{i=1}^n P(A_i)\mathbb{E}[X | A_i].$$

Proof. Substitute the total-probability expression for the PMF into the expectation, then collect terms from each event:

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \sum_{i=1}^n P(A_i)p_{X|A_i}(x) \\ &= \sum_{i=1}^n P(A_i) \sum_x x p_{X|A_i}(x) = \sum_{i=1}^n P(A_i)\mathbb{E}[X | A_i]. \end{aligned}$$

Absolute convergence justifies exchanging the sums. □

Example: *The geometric mean.* Return to the tosses until the first head. Write $m = \mathbb{E}[X]$ and partition by $A_1 = \{X = 1\}$ and $A_2 = \{X > 1\}$. Their probabilities are p and q . On A_1 , $X = 1$. On A_2 , one tail has elapsed, and the remaining tosses start the same experiment. Thus $\mathbb{E}[X \mid X > 1] = 1 + m$. Total expectation gives

$$m = p \cdot 1 + q(1 + m) = 1 + qm.$$

Since $1 - q = p$, solving gives

$$\boxed{\mathbb{E}[X] = \frac{1}{p}}.$$

If $p = 1$, the first toss is a head with certainty and $\mathbb{E}[X] = 1$, which agrees with this formula.

6.6 Joint PMFs, marginals, and conditioning

Two random variables can record different features of the same outcome. To describe them together, assign a probability to each pair of values.

Definition 6.4 [JOINT PMF]. The JOINT PMF of discrete random variables X, Y is

$$p_{X,Y}(x, y) = P(\{X = x\} \cap \{Y = y\}).$$

It is nonnegative and satisfies

$$\sum_x \sum_y p_{X,Y}(x, y) = 1.$$

The pair events are disjoint and cover the sample space, which explains the double sum.

	y	$p_{X,Y}(x, y)$				$p_Y(y)$
	4	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{2}{20}$	0	$\frac{5}{20}$
	3	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{9}{20}$
	2	0	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{1}{20}$	$\frac{5}{20}$
	1	0	$\frac{1}{20}$	0	0	$\frac{1}{20}$
		x				
		1	2	3	4	
$p_X(x)$		$\frac{3}{20}$	$\frac{8}{20}$	$\frac{6}{20}$	$\frac{3}{20}$	

Figure 27: Each cell gives the probability of one pair (x, y) . Column sums give $p_X(x)$ and row sums give $p_Y(y)$. The highlighted row contains the pairs retained when conditioning on $Y = 3$.

To find the probability of $X = x$, allow every possible value of Y :

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

These are the **MARGINAL** PMFs. In the displayed example, the column totals are $(3, 8, 6, 3)/20$, which sum to one. For instance, $p_X(3) = (0 + 3 + 1 + 2)/20 = 6/20$.

If $p_Y(y) > 0$, conditional probability gives

$$p_{X|Y}(x | y) = P(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Here the denominator is the whole row's probability, so

$$\sum_x p_{X|Y}(x | y) = \frac{\sum_x p_{X,Y}(x, y)}{p_Y(y)} = 1.$$

For $Y = 3$, the row has probability $9/20$. Its conditional PMF is

x	1	2	3	4
$p_{X Y}(x 3)$	2/9	4/9	1/9	2/9

Marginalizing adds probabilities over the other variable. Conditioning retains one row and rescales it.

Revision. Conditioning changes the weights in a PMF and in an expectation. Total expectation combines conditional means using event probabilities. The geometric remaining waiting time is memoryless, and its mean is $1/p$. A joint PMF gives pair probabilities; marginals sum across the other variable, while a conditional PMF divides a retained row by its probability.

Common mistakes: replacing $\mathbb{E}[200/V]$ by $200/\mathbb{E}[V]$, leaving retained probabilities unnormalized, forgetting to subtract elapsed tosses in the memoryless statement, or using a conditional PMF when the conditioning event has probability zero.

Check yourself: Why is the conditional mean of the uniform example 3? Why does the geometric recursion contain $1 + m$? How do the joint table's highlighted entries become $2/9, 4/9, 1/9, 2/9$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 6: Discrete Random Variable Examples; Joint PMFs](#), PDF pp. 1–2. The supplied calculations and arguments are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).